COMP1002 – REPORT

INTRODUCTION:

When creating a new office branch, it is very important to carefully consider all the requirements needed in order to have a good cybersecurity and networking infrastructure. In this essay, I will be proposing my solution so that the company can have a fully functioning and effective network for the employees and customers. To make sure my proposal is ideal, I will be coming up with problems the company may face currently or in the future, and how my system will help to solve these issues quickly and effectively. I will be considering the networking architecture as well as the cybersecurity architecture. Security is at the highest priority as the company servers will contain many things such as financial information, customer data and much more.

As we are dealing with a company, security is at the highest priority. The company servers may contain many pieces of vital data such as financial information within the company, customer data and much more. It is important that this information is protected securely and that it is not vulnerable or easy to steal. I will provide my proposal and reasonings behind what I believe the company should and shouldn't have to create an ideal security system. This will help to reduce the risk of data loss, theft, and sabotage [1] – Chris LaBounty.

Another key part to create this ideal system is to make sure that it is fully efficient and that it will meet all the initial requirements. Even though the main point is to create a fully functional system, it is very important to consider other aspects such as if the solution is cost effective too. This would help the company as it would initially be cheaper to create the system, so they can use the money to improve other areas of the company branch to make it better.

SYSTEMS ARCHITECTURE:

It is given in the initial requirements that the new branch system should include – staff PCs, staff Wi-Fi and guest network (admin – 10 hosts, sales team – 28 hosts, guest network Wi-Fi – 20 hosts, employee Wi-Fi – 60 hosts), as well as the company's web presence and e-commerce systems (20 hosts). It is also given that the company plans to increase its sales team further in two years' time. From this I understand that there should be a separate employee and guest Wi-Fi, as well as how many hosts each section will contain. When creating I need top keep in mind to leave spare hosts for the sales team as the company plans to grow that section in the future.

My solution will consider the amounts of hosts and Wi-Fi networks required to create an efficient system that will connect everything together. To create this system, I will need to create a networking architecture that will consider all the hardware and devices required in order to securely create a functional network. After this I will need to create a cybersecurity infrastructure, which will go more in depth and will consider all the software aspects that the system should contain for an ideal company branch system that will meet all the requirements. This means ensuring there are no vulnerabilities that could damage the company in the future. It should securely protect all the data and information kept in the company branch or related to the company.

A computer network is when a set of devices will communicate and share data and resources over the digital interconnections using different protocols. The computer network plays a crucial role in company systems acts like the backbone; this is what ensures that everything is connected to each other so that data packets can be sent from one location to another, it is crucial for the company's efficiency. A good company network solution can be very beneficial for the company, it makes it so that it is possible for you to access data and resources from other devices on the same network, as well as having the option to modify and save it if required. Networking would mean that everything is more efficient, such as devices such as printers being connected automatically rather than manually having to set it up whenever employees need to print something.

A good network will enable the employees to share ideas more easily and work efficiently which can lead to increased productivity, which may lead to more company income [2] – Inspired Techs. Another way networking is important is that it expands the ways that the company can provide their services to customers everywhere.

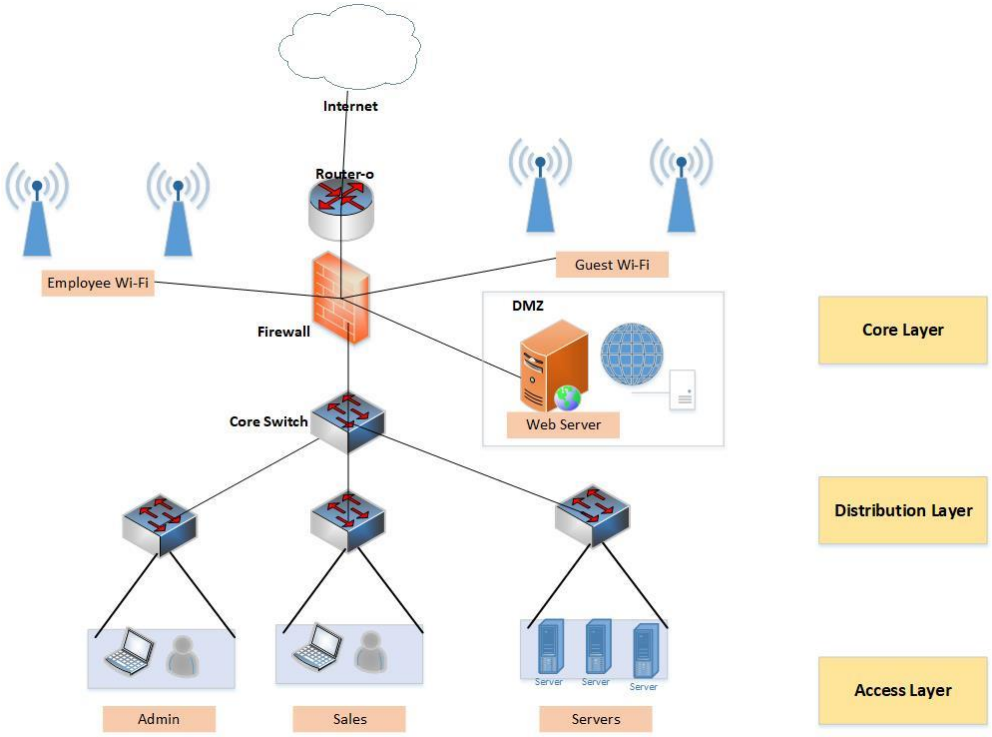
Cybersecurity is also a crucial part when designing the company system. Considering so many employees will work under this branch using company data to complete their tasks throughout the day, as well as having guests using devices under the guest Wi-Fi, security should be at a high priority when considering the company branch's new system. As the company may be working with customers, there is probably a lot of confidential data kept withing servers that should be protected well to avoid data breaches that would lead to data loss or confidential information being leaked, whether it be the company's, employee's, or customers' information. It has been recorded that 46% of UK businesses have faced cyber-attacks (in 2017) [3] – UK GOV. That is almost half of ALL UK businesses! The percentages are decreasing due to the use of more protection to fight against these cyber-attacks. This shows that is important to keep this up so that this company branch will not be susceptible to cyber-attacks either.

NETWORKING ARCHITECTURE:

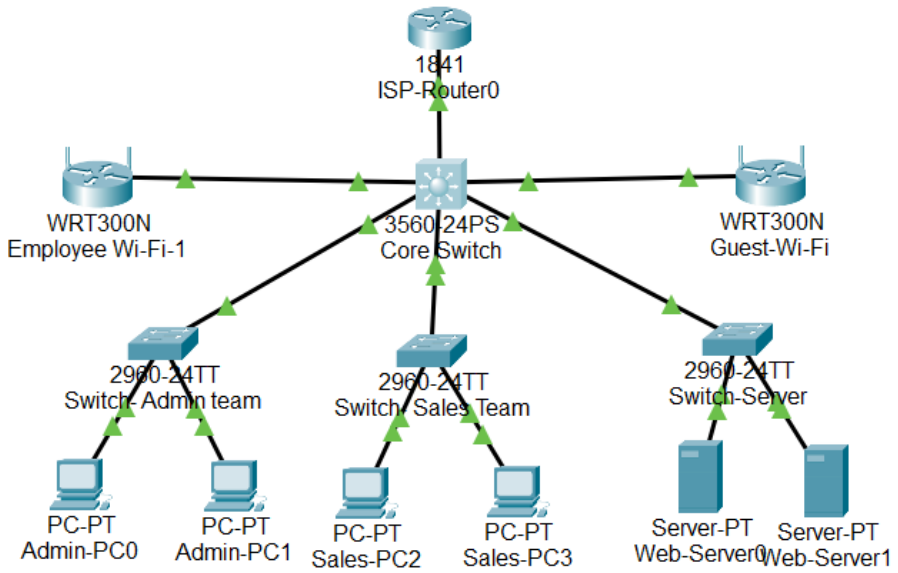
For my networking architecture, I am going to use the hierarchical network design, as I have found that it would be most suitable for this company environment. This is where the network is split up into three distinct layers – the core layer, the distribution layer, and the access layer. The core layer is an essential part which connects different components together and is made up of the high speed, powerful network devices, in the case of my design, the core switch and router (refer to my logical network diagram below). The distribution layer is where the network traffic will be distributed evenly within the network so that high traffic doesn't cause the network to get congested. This is a key part, as without it, data packets would take a very long time to send or get lost while sending; this could be crucial for the company. In this distribution block, I have placed a firewall before the access layer for security as a network policy is usually required here for protection. The access layer

is the last part of this hierarchical design. It is usually considered the entry/exit point to the network because this is where all the different end user devices that require a network connection (such as phones, laptops, PCs, printers…etc) are connected to the network [4] – Auvik Networks.

I have chosen the hierarchical network topology as it seems more efficient compared to other network designs such as the flat network, which is simple and cheap to set up but has very poor security due to everything being connected to one switch. The use of a poor network design like this could cause critical damage to the company. On the other hand, the hierarchical network design has a lot of good points that are very beneficial for the company branch. Firstly, hierarchical networks scale very well; this is good as in the initial requirements is states that they are planning to grow the sales team. Thanks to the scalability, it also means the network is easy to manage and maintain. Another point is that it has very good security, reliability, redundancy and performance which is great for the company [5] – Cisco Training Academy Blog. Due to the design's multiple switches and different pathways, it means that if an access layer switch fails, only that part of the network will be affected, everything else will run fine. Similarly, if a hacker were to somehow gain access to a device and try to take over the system, it is easy for the company to cut off access so that the hacker cannot infiltrate the system any further. The only disadvantage I see within this topology is that if something higher up such as anything in the core layer, such as the core switch were to fail, then the whole system would fail, but this is a very unlikely scenario which is why I think the benefits of this design outweighs the disadvantages. Another possible problem is that when creating this structure, it is easy for the company to spend more than necessary, but this is solved as this design is scalable meaning they can grow it when required, so I believe it is cost effective and worth it.



*The password for the router and switch is 'Cisco123'



The end host devices (PCs and servers) are connected to separate switches, and those switches are connected using a core switch. A switch is a high-speed piece of hardware that connects devices and directs the data packets to their location. The core switch is used for routing and switching data in the core layer of the network. It is not clearly shown it in my layout diagram, but a Virtual Local Area Network (VLAN) is integrated in the switches in the distribution layer. This is like a physical switch which is used to show the end host devices are part of different subnetworks. For example, the admin host and sales host are on different networks. I have placed the web server in a dematerialised zone (DMZ) to create a subnet to separate it from all the other devices using the zero-trust policy as it is better to be safe than sorry. The web server will contain the company's important and confidential information, which means to make it more secure, it would be better to treat all the other devices in the network as untrusted.

The core switch, Wi-Fi networks and web servers are all connected to the router. A router is the device which is responsible in organising communication between all the different devices in the network and the internet. I have placed a hardware firewall between the router and all the devices that it is connected so it can help to inspect the network traffic, alerting the company in case anything is unusual. It can also encrypt and decrypt the information that is being sent/received so that the data packets cannot be stolen easily by hacklers. The company can also access all the unauthorised access attempts and all the other suspicious activities. If there is a growth in this, the company will need to consider another firewall as a precaution, such as between the core switch and the access layer switches.

To create this network, I have implemented Variable Length Subnet Mask (VLSM) subnetting, which has helped me split up this large network into smaller subnets, and subnetting subnets allow us to use multiple masks for the different subnets. In this case it is class C because it is /24, using IPv4 (Internet Protocol version 4) so it is 32 bits available maximum 255. As the company has requested in the initial requirements, I have used more bits than needed in sales so that there would be more space once the team grows in the future.

IP ADDRESS TABLE:

ID = 10745379 X = 7; Y = 9

Network: 79.7.9.0 / 24

Admin

Specification	Value
Number of bits in the subnet	28
New IP mask (decimal)	255.255.255.240
Number of usable subnets	4
No. of usable hosts per subnet	14
Network address	79.7.9.192
First IP Host address	79.7.9.193
Last IP Host address	79.7.9.206

Sales

Specification	Value
Number of bits in the subnet	26
New IP mask (decimal)	255.255.255.192
Number of usable subnets	4
No. of usable hosts per subnet	62
Network address	79.7.9.64
First IP Host address	79.7.9.65
Last IP Host address	79.7.9.126

Guest Wi-Fi

Specification	Value
Number of bits in the subnet	27
New IP mask (decimal)	255.255.255.224
Number of usable subnets	8
No. of usable hosts per subnet	30
Network address	79.7.9.128
First IP Host address	79.7.9.129
Last IP Host address	79.7.9.158

Employee Wi-Fi

Specification	Value
Number of bits in the subnet	26
New IP mask (decimal)	255.255.255.192
Number of usable subnets	4
No. of usable hosts per subnet	62
Network address	79.7.9.0
First IP Host address	79.7.9.1
Last IP Host address	79.7.9.62

Servers

Specification	Value
Number of bits in the subnet	27
New IP mask (decimal)	255.255.255.224
Number of usable subnets	8
No. of usable hosts per subnet	30
Network address	79.7.9.160
First IP Host address	79.7.9.161
Last IP Host address	79.7.9.190

CYBERSECURITY ARCHITECTURE:

The cybersecurity architecture plays a big part in the protection of company information. In 2021, statists have shown that there have been 60,111 cases of cybercrime reported by businesses [6] – Comparitech. This is a very high number, and just goes to show how much extra protection is needed for businesses to take care of their and their customers data. The Company Data Protection Policy states that information of employees, customers and even suppliers should be treated at upmost care and confidentiality [7] – Workable. Multiple layers of security is probably ideal to ensure the system is not vulnerable and open to hackers.

To find out how to protect the system, first let's recognise the common threats and cyber-attacks within businesses. The most common type of threat is malware; this is an umbrella term for other types of threats such as spyware, viruses, and worms. Malware finds a vulnerability within the system in order to breach it, then hides itself in unexpected places within the system. An example of this could be if the malware has planted itself into a link in an email that has been sent to all the employees. If the link is clicked, it will latch onto the device, collect data, and spread it to other devices until the whole system has been infected. This is a very dangerous situation as the collected data could be stolen, used the data for ransom or even delete the data. It has been shown that some 82% of British firms have fallen victims to ransomware attack and have paid the hackers to get their data back [8] – BBC.

In case of this, it would be best to have backup servers with all the information stored on it, but this must extremely secure and should only be accessed by minimal people, such as the admin team. They should also have to complete a multiple step verification to ensure only people who have access can enter. The backup servers are good so that the data is not lost but is not ideal if the data is leaked to other hackers or possibly rivals of the company.

There are a multitude of ways to combat this issue of malware. Firstly, the company should consider creating different accounts for the different people. For example, as there's guest Wi-Fi for guests and employee Wi-Fi for employees, they should create guest accounts for the guests, so they only have a limited access to the company's network. Similarly, if possible, they should control what the employees have access to. For instance, the human resources staff should only have access to data regarding human resources and not finance, and vice versa. Only admins should have access to view more things and possibly modify data if required too. The company should also ensure that the passwords to the accounts are strong; they can do this by creating a criterion to follow when someone sets up their password to enforce and ensure they have a secure password. Even though it is mainly common sense, employees should be advised to not click on unknown links, not to

download unknown files into the company PCs, avoiding opening suspicious email attachments, limiting file sharing…etc.

The company should also install software firewalls, anti-virus, anti-malware, and anti-ransomware software. Having one antivirus programme installed is not enough, especially for a company, and the cost of this protection is not expensive considering how secure the data will be and how it will prevent problems in the future. It would be good to have even more security programmes on the admin PCs as they will have access to more of the data than the regular employee and guest PCs. It might be excessive, but the company could consider using a biometric verification such as an iris or facial scanner for added security.

As well as having all these programmes installed in the PCs for security, it is important that their regular backups are performed, the software is regularly updated and that there are regular scans to detect any sort of suspicious behaviour. Tell the employees to inform the IT team (or admin) if they notice anything strange about the PCs. For example, if the PC is significantly slower than usual, sends emails they don't remember sending, shows unexpected icons/files in your PC…etc, as these are all signs of a PC that is most likely infected with malware. If it is recognised quickly, the issue can be solved at the root of the problem. If it is ignored and the malware grows to the other PCs, or even PCs in other departments, or even servers, it could cause an entire part of the system to fail, which would affect the company and stop business.

It would be best to use HTTPS (with SSL), which is the secure extension of HTTP (Hypertext Transfer Protocol) using TLC (Transport Layer Security)/SSL (Secure Sockets Layer) certificates, where the communication layer of the protocol is encrypted with the focus on increasing the security while transferring sensitive information [9] - CloudFlare. In the company environment, there are many examples of sensitive information, such as employee details, bank information, customer details…etc. It has two different keys, the private and public key, which encrypt and decrypt information. What the private key encrypts can only be decrypted by the private key, and similarly what the public key encrypts can only be decrypted by the public key. The SSL certificates are shown to ensure that the provider is who they say they are, which build trust safely and securely. Without HTTPS encryption, data travelling back and forth from the router and internet is vulnerable and can be easily stolen by hackers, whereas with HTTPS it is encrypted and sent in smaller packets where it is hard to intercept, steal and read.

CONCLUSION:

Carefully considering all the requirements of the Company has asked for its branch, I believe I have built a good networking and cybersecurity infrastructure. By using the hierarchical network topology, the layers will distribute the network traffic efficiently so that there is no congestion in the network that may affect the company. The topology itself as well as the hardware firewall placed in the network will provide essential protection to the system. The hierarchical topology is well known for being stable, as if something were to go wrong within one of the subnets, such as a hacker, only that area (switch and devices) would fail but it would not affect the whole network; this means the damage can be quickly and easily repaired. The firewall allows the company to monitor and filter everything that is being sent and received so that it can detect any suspicious behaviour or unauthorised access attempts. The topology and firewall are cost effective for the company for how well it will work now and, in the future, when teams start expanding. I have also recommended an extra firewall in case there is a high risk of cyberattacks in the future, even though it will be more expensive to have multiple firewalls, it can monitor specific areas of the network so that they can identify and fix the problem areas.

For the cybersecurity, a lot of security software such as anti-malware and anti-virus should be installed, and the company should consider limiting access to particular people such as guests and multifactor authorising people who have more access to the company such as the admins. Through this, as well as the company educating employees about how malware comes about and advising them to not click unknown links or download unknown files on the company PCs, this will ensure that the data is secure and well protected.

In 2020, 46% of UK companies claimed to experience data breaches and cyber-attacks, where 1 in 5 have claimed to lose money, data, or valuable assets. One third of businesses have claimed to be impacted negatively [10] – UK GOV. Over the pandemic this seems to have increased as many companies have been relying on online resources rather than physical, which means it is important to prioritise security in the network and cybersecurity architecture. I believe I have done this well, while looking at other aspects which would be beneficial to the company such as cost, reliability, maintainability, and performance.

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